

EDA SS4 Assignment

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1. What is Exploratory Data Analysis (EDA), and why is it important?

EDA is the first stage of working with a dataset, done before any modelling or formal testing. You use summary statistics (count, mean, median, quartiles, standard deviation) and charts (histograms, box plots, scatter plots, heatmaps) to find out what the data contains and how it behaves.

It is important because it:

- shows the structure of the data: what each row represents, the column types, and how many values are missing;
- reveals problems such as missing values, duplicates, wrong data types and outliers, so they can be fixed or understood before analysis;
- shows the shape of each variable (skewed or symmetric), which decides which statistics and tests are appropriate (for example median instead of mean);
- uncovers patterns and relationships (for example price vs freight value) that suggest good questions and hypotheses.

Skipping EDA can lead to wrong conclusions, such as reporting a mean that is heavily distorted by a few extreme values.

2. What is the difference between a histogram and a bar chart?

Histogram: A histogram shows the distribution of a numerical (continuous) variable. The values are grouped into ranges called bins, and each bar's height is the number of observations in that range. The bars touch because the bins form one continuous number line, and their order is fixed.

Bar chart: A bar chart compares categorical variables. Each bar is a separate category, and its height is a count or a value for that category. The bars have gaps between them, and they can be put in any order (for example sorted from largest to smallest).

In short: a histogram shows how a single number is spread out, while a bar chart compares separate groups.

3. What does `df.head()` do in Pandas?

`df.head()` returns the first 5 rows of the DataFrame `df`. You can pass a number to see a different count, for example `df.head(10)` shows the first 10 rows. It gives a quick preview of the data: the

column names, the kind of values each column holds, and whether the file loaded correctly, without printing the whole dataset.

4. Explain the difference between mean and median.

Mean: The average: add up all the values and divide by how many there are. It uses every value, so it is sensitive to extreme values (outliers).

Median: The middle value once the data is sorted from smallest to largest. If there is an even number of values, it is the average of the two middle values. It is resistant to outliers.

5. What is an outlier? Give one example.

An outlier is an observation that is unusually far from most of the other values in the data. A common rule flags values below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$ as potential outliers. An outlier is not automatically an error. It may be a real but rare case, so it should be investigated before it is removed.

Example: in the Olist data most product prices are between R\$39.90 and R\$134.90, but one item costs R\$6,735.00. The upper fence is $134.90 + 1.5 \times 95.00 = R\277.40 , so the R\$6,735 item is clearly an outlier. It is probably a genuine expensive product, not a mistake.

6. Which chart would you use to investigate the relationship between product price and delivery cost? Explain why.

A scatter plot, with product price on the x-axis and delivery (freight) cost on the y-axis.

- Both variables are numerical, and a scatter plot is the standard chart for showing how two numerical variables relate.
- Each point is one order item, so you can see the direction (positive or negative), the strength and the shape (linear or curved) of the relationship.
- It also shows clusters and outliers, such as very expensive items with low freight.

With a lot of data the points overlap, so I would make them partly transparent (alpha) or use a log scale, and add the correlation coefficient to measure the relationship.

7. What is a heatmap, and what does a correlation close to +1 mean?

A heatmap is a chart that shows a grid of numbers using colour, with darker or warmer colours for higher values. In EDA it is often used to display a correlation matrix: every numerical variable is compared with every other, and the colour of each cell shows the correlation between that pair. This makes strong relationships easy to spot.

A correlation close to +1 means a strong positive linear relationship: when one variable increases, the other almost always increases too, and the points lie close to an upward straight line. Correlation does not prove that one variable causes the other.

8. Understanding Quartiles

Quartiles divide the sorted data into four equal parts, each holding 25% of the observations.

- a) **Q1 (25%):** 25% of the values are at or below Q1 and 75% are above it. It marks the bottom of the middle half of the data. In the Olist data, Q1 = R\$39.90.
- b) **Q2 (50%):** This is the median. Half the values are at or below it and half are above. It is the middle value. In the Olist data, Q2 = R\$74.99.
- c) **Q3 (75%):** 75% of the values are at or below Q3 and only 25% are above it. It marks the top of the middle half of the data. In the Olist data, Q3 = R\$134.90.

d) Why quartiles are useful for product prices:

- Product prices are usually skewed: a few very expensive products pull the mean up. Quartiles are based on position, so outliers barely affect them and they describe a typical product better.
- They show the price range of the middle 50% of products (Q1 to Q3), which is the IQR, a robust measure of spread.
- They are used to detect outliers ($Q1 - 1.5 \times IQR$ and $Q3 + 1.5 \times IQR$) and to draw box plots.
- They help with business decisions, for example grouping products into budget, mid-range and premium price bands.

9. The table below shows the descriptive statistics for product prices in the Olist dataset.(10 mks)

Statistic	Product Price (R\$)
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Mean	120.65
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Standard deviation	183.63
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Minimum	0.85
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Q1 (25%)	30.90
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Q2 (50%)	74.99
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Q3 (75%)	134.90
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Maximum	6,735.00
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Answer the following questions:

9.1 Difference between the mean and median

$$\text{Difference} = \text{Mean} - \text{Median}$$

$$= 120.65 - 74.99$$

$$= R\$45.66$$

The mean is R\$45.66 higher than the median. This shows that a small number of expensive products are pulling the average up.

9.2 Interquartile range

$$\text{IQR} = Q3 - Q1$$

$$= 134.90 - 39.90$$

$$= R\$95.00$$

The middle 50% of products cost between R\$39.90 and R\$134.90, a range of R\$95.00.

9.3 Percentage of products costing R\$134.90 or less

R\$134.90 is Q3, the 75th percentile, so 75% of products cost R\$134.90 or less.

9.4 What the standard deviation tells us

The standard deviation (R\$183.63) measures how far prices typically are from the mean. It is larger than the mean itself ($R\$183.63 > R\120.65), so prices vary a lot: the catalogue ranges from cheap items (R\$0.85) to very expensive ones (R\$6,735). Because the standard deviation is based on the mean, it is also inflated by the extreme high prices. For skewed data like this, the IQR (R\$95.00) gives a more reliable picture of how spread out typical prices are.

9.5 Is the distribution positively or negatively skewed?

Answer: The distribution is positively skewed (right-skewed), with a long tail towards high prices.

Reasons:

- **Mean > median.** The mean (R\$120.65) is greater than the median (R\$74.99). The few very expensive products pull the mean to the right of the middle value.
- **Long upper tail.** The maximum (R\$6,735.00) is R\$6,600.10 above Q3, while the minimum (R\$0.85) is only R\$39.05 below Q1. The upper tail is much longer than the lower tail.
- **Uneven quartile gaps.** The gap from Q2 to Q3 ($134.90 - 74.99 = R\$59.91$) is larger than the gap from Q1 to Q2 ($74.99 - 39.90 = R\$35.09$), so even the middle of the data is stretched to the right.
- **Large standard deviation.** The standard deviation is larger than the mean. Prices cannot go below 0, so this much spread can only come from a long tail of high values.